

Exploratory Data Analysis (EDA): Two Categorical Variables

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MATH 240

Lecture #6 — Ch. 4.2, 4.3

- This week: EDA (conceptually and in R)
- Homework #2 is due **Friday** at 11:59pm
 - ▶ Includes textbook and R problems
 - ▶ Use of `dplyr`, `ggplot2`
- Checkpoint #2 on Friday
 - ▶ Work in groups of 2-3 on EDA tasks using R
 - ▶ 2 sets of problems, each related to a dataset
 - ★ 1 set will be graded for accuracy, the other for completion
 - ▶ Will have the whole period to work on them

Learning Objectives

- Analyze metrics of two or more variables using contingency tables and correlation
- Visualize multiple variables using mosaic plots, faceted plots, box plots, and scatter plots using R

Activity: Analyzing a Single Variable

We will look at the `duke_forest` dataset from the `openintro` package. To load the data in, we will use the following lines of code:

- `library(openintro)`
- `data(duke_forest)`

With the data loaded in, do the following in pairs:

- Create a plot for each of the following variables and describe the distribution's characteristics:
 - ▶ Price
 - ▶ Area
 - ▶ Cooling
- For the aforementioned variables, create appropriate summaries or tables using R.
- For the parking and heating variables: determine how we could transform these variables into meaningful categorical variables.

Movies Dataset

We will continue working with data on movies as a way for us to explore various types of data.

- 3,039 movies made in U.S. between 1929 and 2016, scraped from IMBD (by Dr. Robin Belton)

	movie_title	title_year	budget	log_budget	gross	log_gross
1	Avatar	2009	237000000	19.28357	760505847	20.44949
2	Pirates of the Caribbean: At World's End	2007	300000000	19.51929	309404152	19.55016
3	The Dark Knight Rises	2012	250000000	19.33697	448130642	19.92060
4	John Carter	2012	263700000	19.39032	73058679	18.10677
5	Spider-Man 3	2007	258000000	19.36847	336530303	19.63420
6	Tangled	2010	260000000	19.37619	200807262	19.11786

Figure: First Six Movies in Movies Dataset

Two Categorical Variables: Contingency Tables

- Tool to get counts of categories of multiple categorical variables

Rating	Century		Total
	20th	21st	
G	27	43	70
NC-17	11	2	13
Not Rated	10	21	31
PG	142	329	471
PG-13	192	917	1109
R	460	872	1332
Total	842	2184	3026

Table: Contingency Table of Movies based on Century of Movie Release and Rating

Two Categorical Variables: Contingency Tables

- Joint frequencies

- ▶ Allows for joint proportions

$$\hat{p}_{G,20th} = \frac{\text{joint count}}{\text{grand total}} = \frac{27}{3026} \approx 0.0089$$

Rating	Century		Total
	20th	21st	
G	27	43	70
NC-17	11	2	13
Not Rated	10	21	31
PG	142	329	471
PG-13	192	917	1109
R	460	872	1332
Total	842	2184	3026

Two Categorical Variable: Contingency Tables

- Marginal frequencies

- ▶ Counts for a *single* variable
- ▶ Allows for marginal proportions

$$\hat{p}_G = \frac{\text{marginal count}}{\text{grand total}} = \frac{70}{3026} \approx 0.0231$$

Rating	Century		Total
	20th	21st	
G	27	43	70
NC-17	11	2	13
Not Rated	10	21	31
PG	142	329	471
PG-13	192	917	1109
R	460	872	1332
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Two Categorical Variable: Contingency Tables

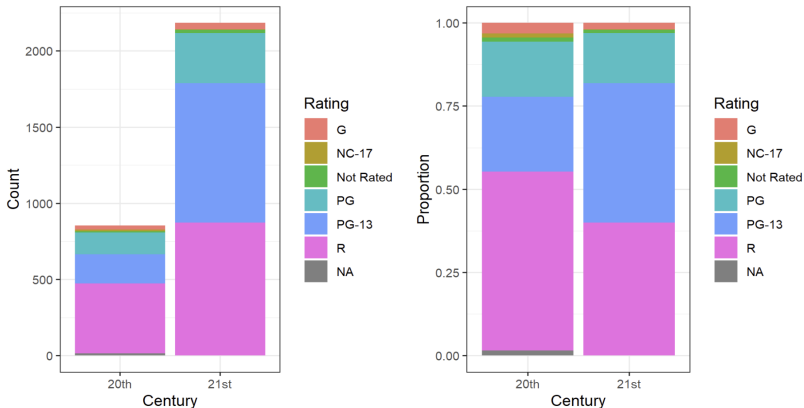
- Joint frequencies
- Marginal frequencies
 - ▶ Allows for conditional proportions

$$\hat{p}_{G|20th} = \frac{\text{joint count}}{\text{marginal total}} = \frac{27}{842} \approx 0.0321$$

Rating	Century		Total
	20th	21st	
G	27	43	70
NC-17	11	2	13
Not Rated	10	21	31
PG	142	329	471
PG-13	192	917	1109
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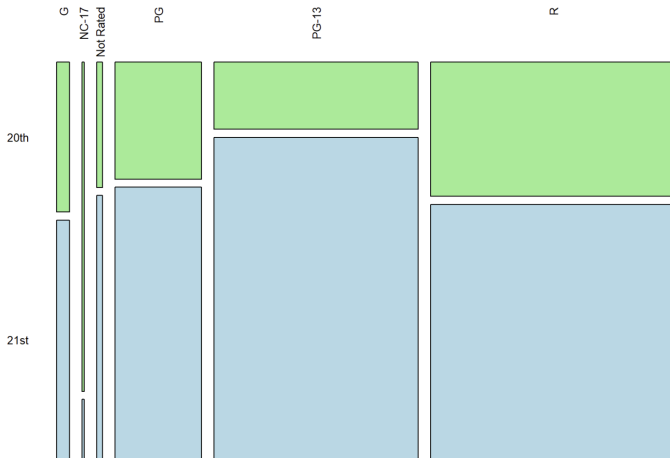
Two Categorical Variable: Stacked Bar Plots

- Allows us to see a distribution of a variable across groups of another variable
- e.g. suppose we want to see the distribution of ratings across time periods



Two Categorical Variable: Mosaic Plots

Mosaic Plot of Content Ratings and Centuries of Movies



Activity: Two Categories

Using the `duke_forest` dataset, do the following:

- Create a new column by doing the following:
 - ▶ Install the `stringr` package
 - ▶ Load the package in R with the `library()` function
 - ▶ Run this code: `duke_forest <- duke_forest |> mutate(gas = str_detect(duke_forest$heating, "Gas"))`
- Create a contingency table of gas and cooling.
- Find the following values:
 - ▶ Joint proportion of houses with no gas and no central cooling.
 - ▶ Marginal proportion of central-cooled houses with gas.
- Create a stacked bar plot of gas and cooling.

For Wednesday!

- Read through chapters 4-5, marking down any questions you may have about material.
- Look through the R code discussed over the last 3 lectures.